

2.10 Cell Signalling

▼ Describe how the body responds to hypoglycaemia.

- Glucagon is released by alpha cells in the Islet of Langerhans (pancreas)
- Glucagon travels from pancreas to blood vessels
- This stimulates **glycogenolysis** (breakdown of glycogen) and **gluconeogenesis** (generation of glucose) in the liver, increasing blood glucose levels.

▼ What is meant by "endocrine signalling"?

- This describes how **secreted hormones travel from one tissue** in the body **to another distant target cell** via the bloodstream
- For example, **insulin** is secreted by the **pancreas** to act on the **liver**.

▼ Describe how the body responds to hyperglycaemia.

- Insulin is released by beta cells in the Islet of Langerhans (pancreas)
- This **prevents further glucagon secretion in the pancreas**, preventing further increase of blood glucose levels.

▼ What is meant by "paracrine signalling"?

- The action of a **hormone** on an **adjacent target cell**.

▼ Recall two other examples of paracrine signalling.

- **Nitric oxide** produced by **endothelial cells in blood vessels**, acting on **smooth muscle cells** in blood vessels to cause **vasodilation**
- **Osteoclast** activating factors produced by adjacent osteoblasts (bone remodelling).

▼ Describe an example of signalling between membrane attached proteins.

- **Blood-borne virus** detected by **antigen-presenting cell (APC)**
- **APC** digests pathogen and expresses MHC Class II molecules on its surface
- This leads to T Cell receptor (**TCR**) interacting with this MHC and **binding to the antigen**

▼ What is meant by autocrine signalling?

- **Signalling molecules being released** by one cell and **binding onto the same cell, acting on it**.

- ▼ Describe how the process of neurotransmission between the presynaptic and postsynaptic axon terminals occurs.
 - The **action potential is propagated** by the **depolarisation** (Na⁺ influx) and **repolarisation** (K⁺ efflux) of the **presynaptic membrane**
 - The **action potential leads to Ca²⁺ influx** into the presynaptic axon terminal and **exocytosis**
 - The **neurotransmitter binds** to the **postsynaptic receptors**
 - The **postsynaptic receptors** are activated and **signal is transmitted**.
- ▼ Where on cells are postsynaptic receptor molecules located?
 - On cell membranes
- ▼ Recall the four types of postsynaptic neurone receptor.
 - Ionotropic receptor
 - G-protein coupled receptor
 - Enzyme-linked receptor
 - Intracellular receptor
- ▼ What is meant by the term "ligand" in cell signalling?
 - A hormone, protein, molecule
 - Just anything that can bind to a receptor
- ▼ Describe how ionotropic receptors work.
 - **Ligand binds** onto the **receptor**
 - This causes an **ion channel protein to open** a pore in the cell membrane
 - The **pore causes ions to travel in and out of cells** depending on the **concentration gradient**.
- ▼ What is an alternative name for a G-protein coupled receptor and why is that name given to it?
 - 7-TM (transmembrane) receptor
 - This is because it crosses the membrane seven times.

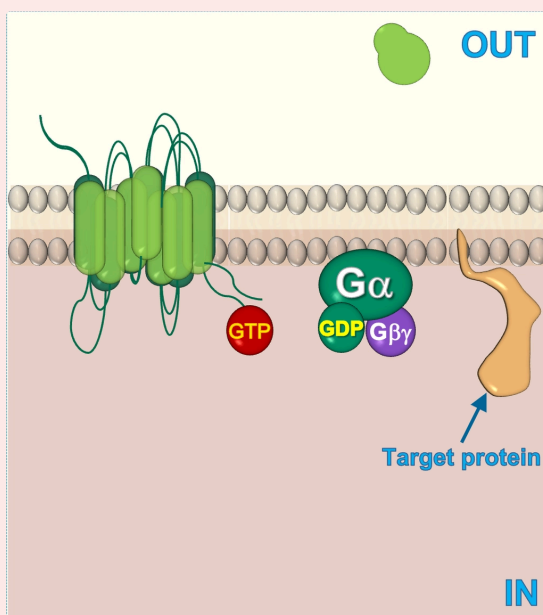
▼ What are the three constituents present in a G protein?

- **Alpha**
- **Beta** and **gamma** which stay together
- **GDP** is also bound to this complex



▼ Describe how G-protein coupled receptors work.

- **Ligand binds onto the receptor** which **activates** it and causes it to **change conformation**
- This allows the previously unassociated **heterotrimeric G-protein** to **bind to the receptor**
 - **NOTE: multiple G-proteins bind to the receptor as a result of one G protein binding**
- **GDP** molecule is exchanged for a **GTP** molecule and the G-protein is activated
- The **G-protein** then **separates** into its **alpha** and the **beta-gamma** subunits
- The **alpha subunit**, attached to the **GTP** molecule, then moves to the **target protein**
 - **GTP is dephosphorylated to GDP** using **internal GTPase**, this energy is used to move the **alpha-GTP** complex to the **target protein**
- Effects are then induced depending on the **identity of the target protein**
- The **alpha subunit** then **rebinds** to the **beta and gamma subunits** and **reforms the G-protein**
- The **process stops** when the **ligand is no longer bound** to the **cell surface**



▼ What are the two most prominent types of G-protein?

- Gs and Gi.
- s for **stimulate** (adenylyl cyclase activity), i for **inhibit** (adenylyl cyclase activity).

▼ What is the third type of G-protein?

- Gq.

▼ What is adenylyl cyclase?

- This is **one of the target proteins that G proteins can act on** during the cascade of reactions

▼ Which G proteins act on adenylyl cyclase and what effects do they have?

- **Gs** - stimulates adenylyl cyclase expression
- **Gi** - inhibits adenylyl cyclase

▼ What does adenylyl cyclase do?

- It stimulates conversion of **ATP to cAMP (cyclic adenosine monophosphate)**
- **cAMP stimulates protein kinase A production**
- This **protein kinase A (PKA)** has various intracellular effects within cells.
- i.e increased protein kinase A can be used to release adrenaline

▼ Recall two examples of G-protein coupled receptors.

- Beta1-adrenergic receptor
- M2-muscarinic receptor

▼ What is the beta1-adrenergic receptor used for?

- **Adrenaline binding**
 - This can induce effects such as **increasing heart rate**

▼ What is the M2-muscarinic receptor used for?

- **Acetylcholine binding**
 - This can induce effects such as **decreasing heart rate**

▼ What does the Gq protein do?

- Stimulates **phospholipase C**
- Converts **PIP to IP3 and DAG**
- **IP3 causes Ca²⁺ release** which is useful for exocytosis
- **DAG activates PKC (protein kinase C)**

▼ How does an enzyme-linked receptor work?

- **At least two ligands bind to the receptor** which causes **clustering of receptors**
- The **clustering activates the enzymes** that are linked to the receptors
- The **active enzymes then phosphorylate** the receptor
- **Phosphorylation** leads to the **binding of signalling proteins** to the cytoplasmic domain of the receptor
- These **signalling proteins recruit other signalling proteins** and **signal is generated throughout the cell.**

▼ Recall two examples of enzyme-linked receptors.